# Iterative programming

# Iterative programming

The concept of iteration underlies any operation in programming languages. In R many operations are vectorized. This makes the language more efficient and cleaner BUT hides the concept of iteration.

Example: if I ask you to use the **`print()`** function to write “hello world” in the console 5 times, how would you do it?

```
msg <- "hello world"
print(msg)
```

```
[1] "hello world"
```

```
print(msg)
```

```
[1] "hello world"
```

```
print(msg)
```

```
[1] "hello world"
```

```
print(msg)
```

```
[1] "hello world"
```

```
print(msg)
```

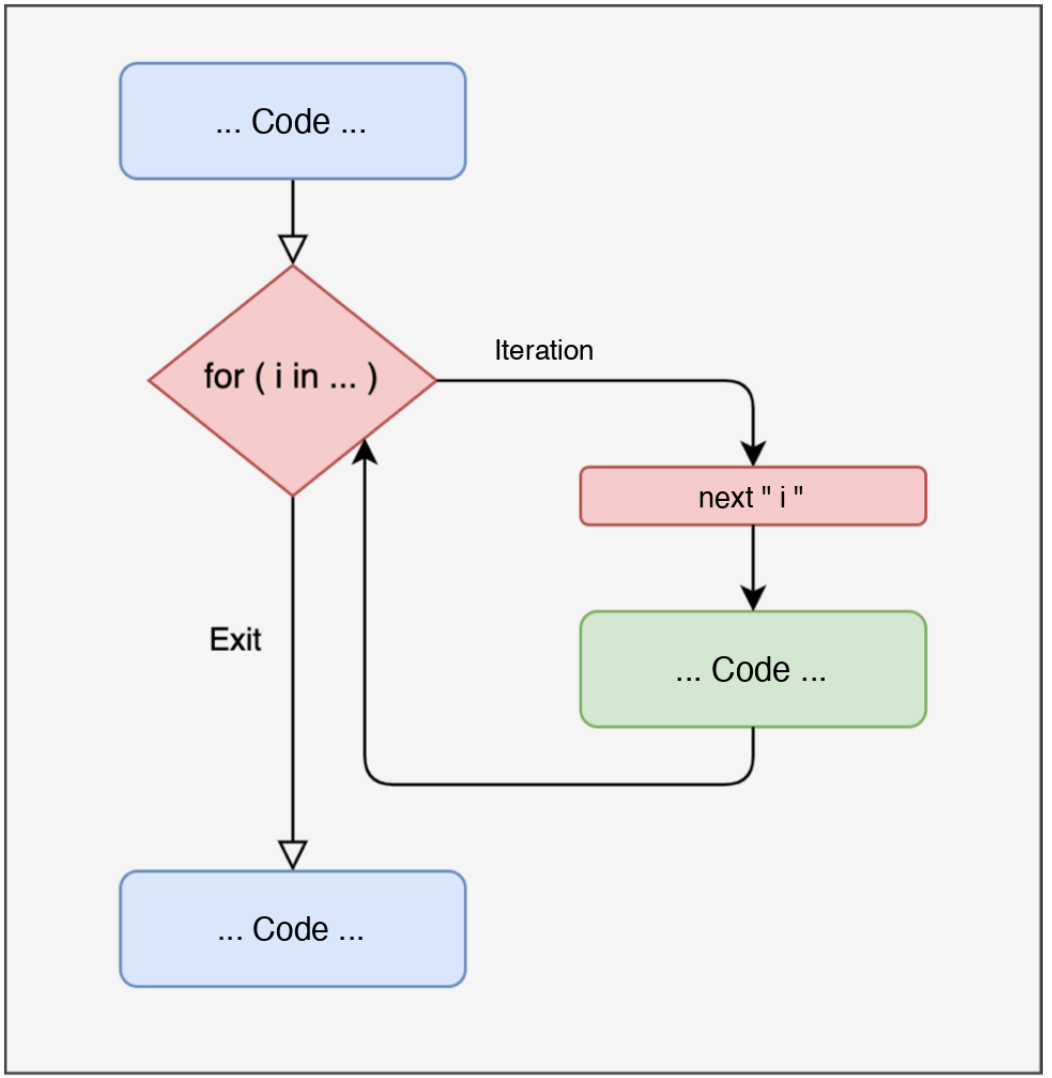
```
[1] "hello world"
```

What we are missing is a way to repeat a given operation without actually repeating the code manually.

There are several constructs that let us repeat operations; the most used are:

- The `for` loop
- The `while` loop
- The `repeat` loop
- The `*apply` family

# The `for` loop



The `for` loop is a structure that lets us repeat a portion of code a finite, pre-determined number of times.

A for loop is written as:

```
n <- 10
for(i in 1:n){
# which operations
}
```

If I want to print something 5 times, I can simply use a for loop:

```
for(i in 1:5){
print(paste("hello world", i, sep = "_"))
}
```

```
[1] "hello world_1"
[1] "hello world_2"
[1] "hello world_3"
[1] "hello world_4"
[1] "hello world_5"
```

# Breaking down the `for` loop

- `for(){ }`: this is the implementation in R (similar to the if statement)
- `i`: this is called the iterator or index. It is a generic index that can take any value and name. By convention it is called i, j etc. It keeps track of the number of iterations our loop has to do
- `in <values>`: this indicates the values the iterator will take inside the loop
- `{ # operations }`: these are the operations the loop must perform

The power of the for loop lies in the fact that the **iterator** i takes the values of the vector specified after `in`, one at a time:

```
for(i in 1:10){
  print(i)
}
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
```

# `For` with an index vs without

`i` represents the index of the elements of the vector vec

```
1  vec <- 1:5
2
3  for(i in 1:length(vec)){
4
5    print(vec[i])
6
7    }
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
```

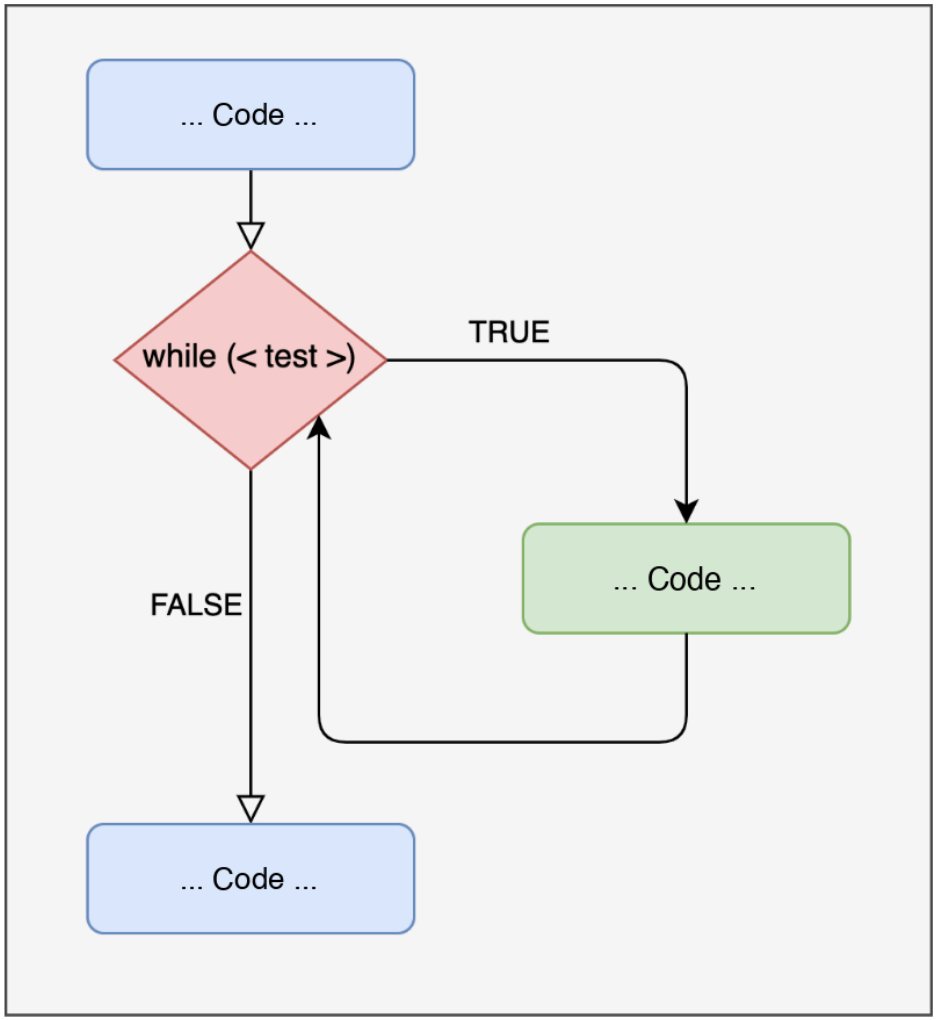
`i` directly represents each element of the vector vec

```
1  vec <- 1:5
2
3  for(i in vec){
4
5    print(i)
6
7  }
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
```

# The `while` loop

The `while` loop uses a logical condition instead of an iterator and a range of values as in the for loop. The loop keeps going as long as the condition is `TRUE`:



Try writing this **while** loop, see what happens and understand why

```
x <- 10

while (x < 15) {

  print(x)

}
```

The **while** loop is not pre-determined, so it always needs a way to be stopped, by making the condition false.

```r
x <- 5

while (x < 15) {
  print(x)
  x <- x + 1
}
```

```
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
[1] 11
[1] 12
[1] 13
[1] 14
```

# The `repeat` loop

The logic of the `repeat` loop is very similar to that of the `while` loop, but with 3 important differences:

1. it always performs at least one iteration
2. it emphasizes repetition until a condition is met
3. it uses the `break` command to stop

# The `repeat` loop

Roll the die until a 6 comes up:

```
repeat{
  die_roll <- sample(1:6, 1)
  print(die_roll)
  if(die_roll == 6) break
}
```

```
[1] 5
[1] 3
[1] 1
[1] 1
[1] 2
[1] 5
[1] 1
[1] 2
[1] 3
[1] 5
[1] 4
[1] 1
```

# `repeat` vs. `while`

`repeat` evaluates the condition at the **end** of the iteration, while `while` does it at the **beginning**. If the condition is not `TRUE` at the start, while does not run, whereas repeat does:

```
1 i <- 1
2 repeat {
3   print(i)
4   i <- i+1
5   if(i > 3)break
6   }
```

```
[1] 1
[1] 2
[1] 3
```

```
1 i <- 1
2 while (i < 4) {
3   print(i)
4   i <- i+1
5   }
```

```
[1] 1
[1] 2
[1] 3
```

# Example: the `sum()` function

Imagine we did not have the `sum()` function and wanted to recreate it: how would we do it? Any ideas?

# Sum as iteration

Let’s break the sum down conceptually; we add up the numbers from 1 to 10:

- I take the first one and add it to the second (total = 1 + 2)
- I take the total and add it to the 3rd element: total = total + 3 …

In practice we have:

- our vector to sum
- a total object that progressively accumulates the previous sums

# Sum as iteration

```r
total <- 0 # initialize the total to 0
x <- 1:10 # vector x

for(i in 1:length(x)){ # for the whole length of x = 1:10
  total <- total + x[i]
}
```

Let’s put everything inside a function (i.e., I create a function that replicates what the sum function does)

```r
my_sum <- function(x){
  total <- 0 # initialize the total to 0
  for(i in 1:length(x)){  # equivalent to 1:length(x)
    total <- total + x[i]
    }
  return(total)
}

x <- rnorm(100, mean = 0, sd = 1)

my_sum(x)
```

```
[1] -0.665768
```

```r
sum(x)
```

```
[1] -0.665768
```

# Iteration, applications

The `for` loop is also useful, for example, to simulate data and perfom a power analysis. Suppose we have a variable y (income) that depends on age:

1. Load the dataset:

```r
mydf <- read.csv(file = "data/mydf_2.csv") # load the dataset

str(mydf)
```

```
'data.frame':   30 obs. of  4 variables:
 $ id     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ age    : int  36 29 37 49 33 50 16 24 48 46 ...
 $ age_cat: chr  "adult" "young" "adult" "adult" ...
 $ age_z  : num  0.27244 -0.35627 0.36226 1.44004 0.00299 ...
```

2. Simulate the variable **y** (income) assuming it depends on **age_z** (age), so we need to extract age_z and choose an effect size

```
# Observed data
age_z <- mydf$age_z
eff_age <- .2 # age effect
nobs <- nrow(mydf)
```

3. Now the for loop …

```r
set.seed(111) # for reproducibility
#initialize the empity vectors
niter <- 1000 # number of iterations
res_all <- rep(NA, niter)    # all estimated effects
p_val <- rep(NA, niter)      # all p-values

for(i in 1:niter) {

  y <- rnorm(nobs, mean = 0 + eff_age * age_z, sd = 1)

  # linear model
  mod <- lm(y ~ 1 + age_z)

  # extract the effect estimate
  res_all[i] <- coef(mod)[2]
```

Result:

```r
# Keep only significant results
res_sig <- res_all[p_val <= 0.05]

# Descriptive statistics
cat("Proportion significant = Power:", mean(p_val <= 0.05), "\n")
```

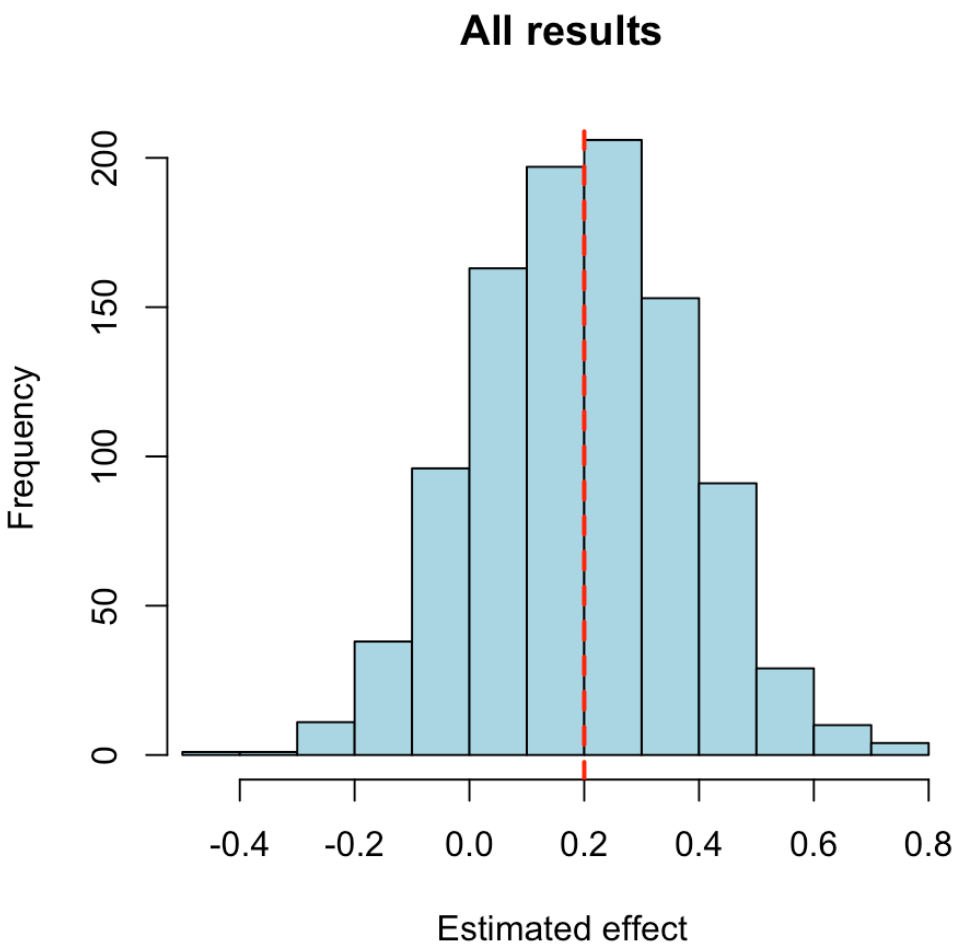
```
Proportion significant = Power: 0.176
```

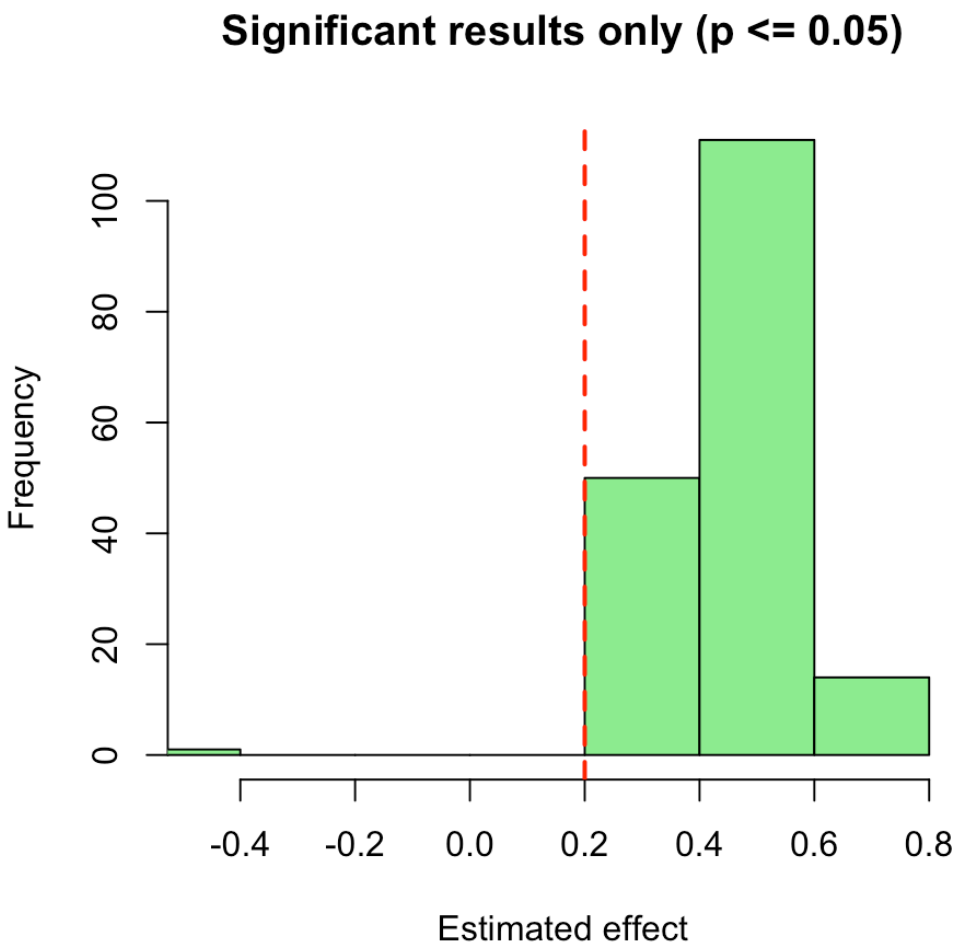
```r
cat("Mean of all effects:", mean(res_all), "\n")
```

```
Mean of all effects: 0.1963144
```

```r
cat("Mean of significant effects:", mean(res_sig), "\n")
```

```
Mean of significant effects: 0.450603
```


All results
Frequency
0
50
100
150
200
-0.4
-0.2
0.0
0.2
0.4
0.6
0.8
Estimated effect


Significant results only (p <= 0.05)
Frequency
0
20
40
60
80
100
-0.4
-0.2
0.0
0.2
0.4
0.6
0.8
Estimated effect

# Nested `for` loops

Once you understand the iterative structure, you can easily extend it by putting a loop inside another one:

```
for(i in 1:3){ # level 1
  for(j in 1:3){ # level 2
      print(paste(i, j))
   }
}
```

```
[1] "1 1"
[1] "1 2"
[1] "1 3"
[1] "2 1"
[1] "2 2"
[1] "2 3"
[1] "3 1"
[1] "3 2"
[1] "3 3"
```

Two groups, **Control** and **Treatment**, with 3 participants each: a nested loop goes through **every participant of every group**.

```r
groups <- c("Control", "Treatment")
for (group in groups) {      # outer loop: groups
  for (subj in 1:3) {        # inner loop: participants
    print(paste("Group:", group, "- Participant:", subj))
  }
}
```

```
[1] "Group: Control - Participant: 1"
[1] "Group: Control - Participant: 2"
[1] "Group: Control - Participant: 3"
[1] "Group: Treatment - Participant: 1"
[1] "Group: Treatment - Participant: 2"
[1] "Group: Treatment - Participant: 3"
```

For each group, the inner loop runs **completely**: 2 x 3 = 6 repetitions.

Now let's **simulate** a reaction time (in ms) for each participant, and save it in a matrix: one **row** for each participant, one **column** for each group.

```
rt <- data.frame(Control = c(NA,NA, NA),
                 Treatment = c(NA,NA, NA))   # empty df
rt
```

```
  Control Treatment
1      NA        NA
2      NA        NA
3      NA        NA
```

The two indices of the loops are the **row** (`i`) and the **column** (`j`) of the cell to fill: `rt[i, j]`

```r
set.seed(42)            # reproducibility
means <- c(500, 450)    # Control, Treatment

for (j in 1:2) {        # columns: groups
  for (i in 1:3) {      # rows: participants
    rt[i, j] <- round(rnorm(1, mean = means[j], sd = 50))
  }
}

rt
```

```
  Control Treatment
1     569       482
2     472       470
3     518       445
```

```r
colMeans(rt)            # mean RT of each group
```

```
  Control Treatment
 519.6667  465.6667
```

# Now let’s practice!

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https://etherpad.wikimedia.org/p/arca-corsoR